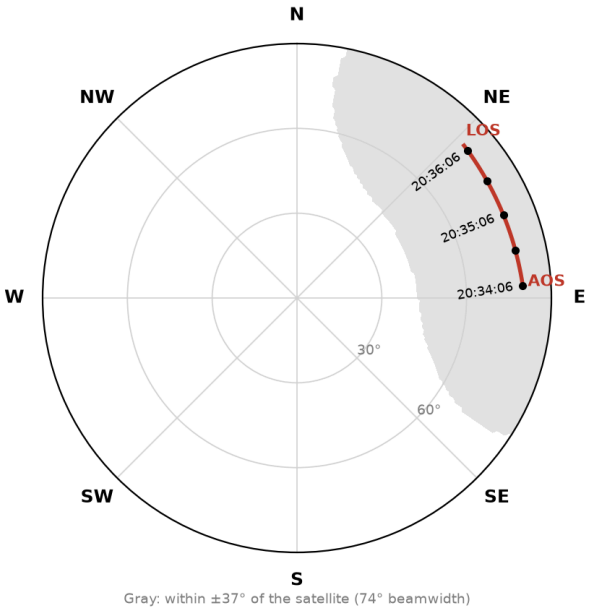
# LilacSat-2 Pass Plan

**Sunday 10/18/2026** Rise 20:34:06 Max 20:35:08 Set 20:36:11 Max El 11° Az 87°→48°

**TX (uplink) 144.350 MHz FM, no tone** **RX (downlink) 437.200 MHz FM** — Doppler-corrected, rounded to 5 kHz (error up to 2.5 kHz)

| Time | TX MHz | RX MHz | Az | El |
|---|---|---|---|---|
| 20:34:06 | 144.350 | 437.205 | 87° | 10° |
| 20:34:36 | 144.350 | 437.200 | 78° | 11° |
| 20:35:06 | 144.350 | 437.200 | 68° | 11° |
| 20:35:36 | 144.350 | 437.200 | 58° | 11° |
| 20:36:06 | 144.350 | 437.195 | 49° | 10° |



Gray: within ±37° of the satellite (74° beamwidth)

**Yaesu VX-6R + speaker mic: uplink TX**

- **Set Mode access:** [F/W] then [0(SET)], dial to the item, [0(SET)] to edit, dial = value, PTT = save
- **Frequency:** VHF band, VFO: key in digits for 144.350; the dial retunes for Doppler
- **Tone:** none: [F/W] then [MODE(SPS) SQ TYP], dial until no TONE shows, PTT to save
- **Power:** [F/W] then [7(LK)TXPO] cycles HIGH / LOW3 (2.5 W) / LOW2 (1 W) / LOW1; use LOW3
- **Set Mode 1 APO:** OFF
- **Set Mode 4 ARS:** OFF
- **Set Mode 6 BCLO:** OFF, so PTT is never blocked by the downlink
- **Set Mode 27 HLF.DEV:** OFF: full deviation
- **Set Mode 37 MCGAIN:** LVL 5, adjust to your voice on the speaker mic
- **Set Mode 51 RPT:** SIMP (simplex); shortcut [F/W] [6(RPT)]
- **Set Mode 67 TOT:** 1 min (default 3)
- **Set Mode 70 TXSAVE:** OFF
- **Do not:** tap [7(LK)TXPO] alone: it turns on the Internet DTMF beep at every TX

**Alinco DJ-VX50: downlink RX**

- **Menu access:** FUNC, then 8, UP/DOWN to the menu number, dial = value, FUNC = confirm, ESC = exit
- **Frequency:** VFO mode: key in digits, e.g. 437.200 = 4 3 7 2 0 0; the dial then retunes for Doppler
- **Menu 01 R-CDC:** OFF (default): a tone squelch would mute the downlink
- **Menu 03 STEP:** 5K (default is 10K), so each dial click is one ladder step
- **Menu 04 W/N:** WIDE (default)
- **Menu 17 SAVE:** OFF (default is ON): battery saver can clip weak openings
- **Menu 19 SQL:** start at 1-2 (default 9 is tight); raise only if noise is constant
- **Menu 20 REPEAT:** OFF (default), and FUNC 4 until no offset is shown: simplex
- **Keypad lock:** FUNC, then hold #; the dial still retunes, and PTT stays untouched on this radio
- **Audio:** Earphone, volume up enough to hear a weak FM carrier

**Notes**

- FM transponder planned for 24 h from ~22:00 UTC Mon/Wed/Fri (AMSAT: flexible, subject to change).
- Listen on the downlink first; a telemetry beacon may be heard when the transponder is off.

**General**

- **Identify:** KF8EEZ, short calls; low power. Keep both radios and the TX antenna well away from the SDR
- **Two radios:** Yaesu VX-6R + MH-57A4B speaker mic = uplink TX (log-periodic); Alinco DJ-VX50 = downlink RX. Use an earphone on the Alinco so its speaker doesn't feed the speaker mic
- **Aim (log-periodic):** point AT satellite: pre-aim at rise Az, follow the Az/El ladder; slant 45°